

Particles and Nuclei 2

- 2/24/26
- Announcements:
- HW 3 Due in One Week
- Reading: (Right after class?) Optional Griff Section 9.5.
Griff Ch 6.

Outline

- Couple more topics from Ch 4
- A couple more short points about isopin
- CP Violation → Weak Interactions generally
- Baryon Number
 - Ch 5 (brief)
- Ch 6

This is where Chapter 9, section 9.5 in Griff's comes into play, we will cover it again but good to get a chunk of the many complications in that chapter out of the way early.

Isospin – in QFT

- From L&B new reading – interesting : deriv. of the isopin conserved current (for triplet—using $SO(3)$ sym – Rotation)
- Similar to conserved “charge” $U(1)$ symm result but diff combo of annih/creation

$$J_N^{3\mu} = \Pi^{a\mu} D^3 \Phi^a = (\partial^\mu \phi^1) \phi^2 - (\partial^\mu \phi^2) \phi^1. \quad (13.14)$$

Swapping signs over, the conserved charge is given in terms of (normally ordered⁶) creation and annihilation operators by

$$\hat{Q}_{Nc}^3 = -i \int d^3p \left(\hat{a}_{1p}^\dagger \hat{a}_{2p} - \hat{a}_{2p}^\dagger \hat{a}_{1p} \right). \quad (13.15)$$

Rotations about the 1- and 2-axes will lead to similar conserved charges in terms of the other field operators. In general the conserved charge will be given by

$$Q_{Nc} = \int d^3x (\Phi \times \partial_0 \Phi) \quad \text{and} \quad \hat{Q}_{Nc}^a = -i \int d^3p \epsilon^{abc} \hat{a}_{bp}^\dagger \hat{a}_{cp}. \quad (13.16)$$

This conserved charge is isospin. As shown in Exercise 13.1, isospin has the structure of angular momentum: in this case an isospin-1. Remember that, despite the similarity between the angular momentum in Exercise 3.3 (which arises since the symmetry is with respect to spatial rotations) and the conserved charge in this example, the symmetry here is *internal* and so this angular momentum-like charge is internal to the field.

To be compared w/ $U(1)$ result here:

Of course, there's the ordering ambiguity here, but we'll press on and insert the mode expansion to yield

$$\hat{Q}_N = \frac{1}{2} \int d^3p \left(-\hat{a}_p^\dagger \hat{a}_p + \hat{b}_p \hat{b}_p^\dagger - \hat{a}_p \hat{a}_p^\dagger + \hat{b}_p^\dagger \hat{b}_p \right). \quad (12.13)$$

Finally, we normal order to give a conserved charge of

$$N[\hat{Q}_N] = \int d^3p \left(\hat{b}_p^\dagger \hat{b}_p - \hat{a}_p^\dagger \hat{a}_p \right). \quad (12.14)$$

Total Conservation

	Q	L	B	J	P	G	C	CP	T	CPT
Strong	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Electromagnetic	✓	✓	✓	✓	✓	X	✓	✓	✓	✓
Weak	✓	✓	✓	✓	X	X	X	X	X	✓

Q = electric charge, L = lepton number, B = baryon number, J = spin,
P = parity, G = G-parity, C = charge conjugation, T = time reversal

CP, CPT are products

Not included....quark flavour...which is conserved by the strong and electromagnetic interaction and by the neutral weak interaction but not by the charged weak interaction. Also, could add isospin I and third component of isospin I_3 (also, did not include cons. of energy etc).

We will discuss CP, T, and CPT next time.

CP Violation – Matter/AM Asymm

- Antimatter vs Matter distinguisher

$$(a) \pi^+ + e^- + \bar{\nu}_e \quad \text{or} \quad (b) \pi^- + e^+ + \nu_e$$

$$\frac{BR(K_L^0 \rightarrow \pi^- e^+ \nu_e) - BR(K_L^0 \rightarrow \pi^+ e^- \bar{\nu}_e)}{BR(K_L^0 \rightarrow \pi^- e^+ \nu_e) + BR(K_L^0 \rightarrow \pi^+ e^- \bar{\nu}_e)} = .0033$$

Notice –
an “A”
form

- Matter/Antimatter Asymmetry of Universe ?
- CP Violation not big → complicated ways to explain
 - have imagined ways cause neither are other parts of the Big Bang understood (e.g, Inflation)
- Heavy Ion Collisions – Asymm generated by the QGP?
 - Originally one minor goal of checking – QGP results in total symmetry matter antimatter (correcting for original matter baryons)
- Regions of Antimatter—more Anti matter?
 - AMS Experiment

C/CP for Weak Interaction/Strong

$$C|v_L\rangle \rightarrow |\bar{v}_L\rangle \quad \text{No!}$$

We have already stated that all anti-neutrinos are right-handed so this is an unphysical state. So charge conjugation invariance cannot be respected by the weak interaction.

Note though that the combined operations of charge conjugation and parity inversion take a left handed neutrino into a right-handed anti-neutrino

$$CP|v_L\rangle \rightarrow |\bar{v}_R\rangle \quad \checkmark$$

(the spin of the neutrino does not transform, but the velocity vector used to define the helicity does).

CP is NOT conserved by only weak –however, “just barely” not per mille effect

$$CP|v_L\rangle = |\bar{v}_R\rangle + \sim 0.001 |\bar{v}_L\rangle$$

RECALL: CKM (P-NMS) Matrix - Oscillations

- Weak Interaction States $\neq$ Mass "Eigenstates"
 - ("What is a particle"?)
- Many effects: E.g. any weak process that produces "one" weak flavor (e.g. " ν_μ ") produces mixed mass eigenstates ($\nu_\mu + \nu_e \dots$)
 - Neutrinos: "free particles" -- Time oscillation phase of "wave function components" go with mass eigenstates ... hence "neutrino oscillations"
 - For quarks, always in bound states, but "oscillations" still seen e.g. in alternate decay modes
- Matrix -- Unitary I think this is main constraint against fourth generation, though some of the measured numbers may assume unitarity
 - CKM: Cabibbo-Kobayashi-Maskawa - quarks

$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix}$$

CKM Strongly diagonal

$$\begin{bmatrix} 0.97370 & 0.2245 & 0.00382 \\ 0.221 & 0.987 & 0.0410 \\ 0.0080 & 0.0388 & 1.013 \end{bmatrix}$$

remember effects will with the squares of these numbers

- P-NMS: not as diagonal
 - Pontecorvo-Maki-Nakagawa-Sakata - neutrinos

$$|U| = \begin{bmatrix} |U_{e1}| & |U_{e2}| & |U_{e3}| \\ |U_{\mu 1}| & |U_{\mu 2}| & |U_{\mu 3}| \\ |U_{\tau 1}| & |U_{\tau 2}| & |U_{\tau 3}| \end{bmatrix} = \begin{bmatrix} 0.801 \dots 0.845 & 0.513 \dots 0.579 & 0.143 \dots 0.156 \\ 0.233 \dots 0.507 & 0.461 \dots 0.694 & 0.631 \dots 0.778 \\ 0.261 \dots 0.526 & 0.471 \dots 0.701 & 0.611 \dots 0.761 \end{bmatrix}$$

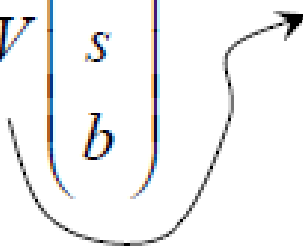
Clarification: CKM $\neq$ CP Violation

LAST TIME

- CKM (NMS/ ν) “mixing” is not really the “cause” of CP Violation or vice versa
 - Mixing Necessary for CP vio, but not “driver”, CP vio doesn’t affect mixing generally (exceptions)
 - Mixing is determined by modulus’s of CKM matrix elements $\rightarrow$ determines “oscillations” e.g. like for ν_e
 - Amount of CP violation: a phase only in some CKM terms (mod 1), only detected through INTERFERENCE btw 2 processes

Weak Force
Lagrangian term

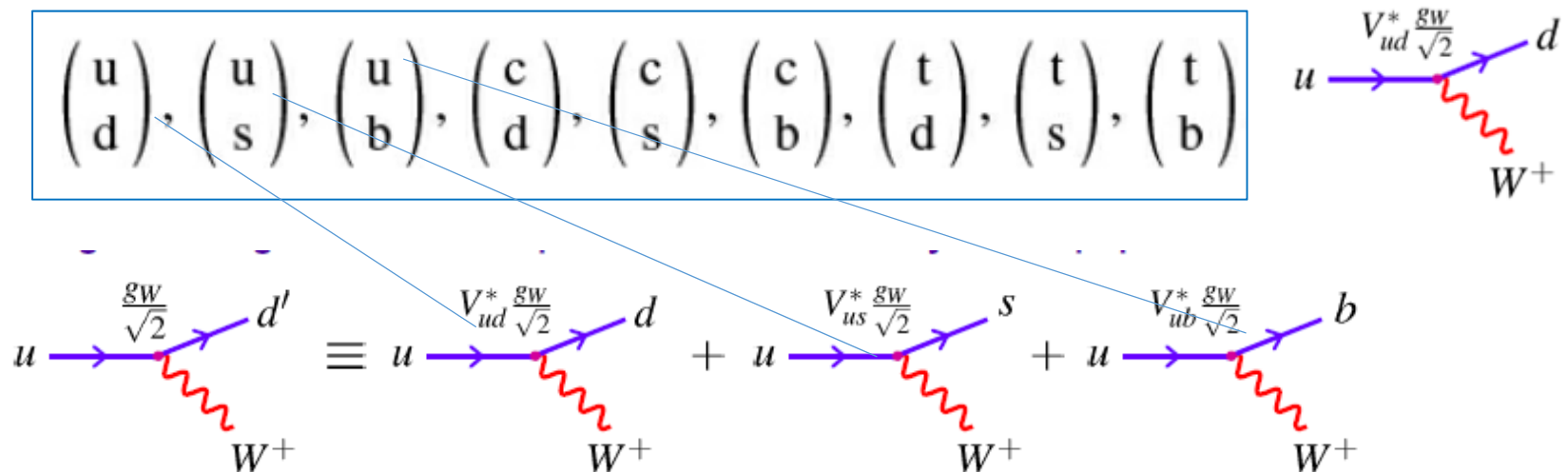
$$(\bar{u}, \bar{c}, \bar{t}) \gamma^\mu (1 - \gamma^5) V \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$



$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix}$$

Reminder Quark Ch Weak Vtx/CKM Mixing

- Remember vertex changes flavor, by “destroying/recreating” it (W carries no flavor $\rightarrow$ only charge) and any q w/charge can be used
- Charged weak vertex possibilities:



- Thomson: starts listing all 9 in Ch 1 intro. Later explains connection to V_{ud}
- Note : confusing terminology :
 - coupling only “3 couplings” no e.g. u to s'
 - d' to u , ... etc. OK $\rightarrow$ no u' , s' , u' to d' (no prime prime vtx)
 - W “creates” d' from u historically influenced choices
- CKM Matrix : defines relative probs of these pairings

$$\begin{pmatrix} u \\ d' \end{pmatrix}, \begin{pmatrix} c \\ s' \end{pmatrix}, \begin{pmatrix} t \\ b' \end{pmatrix}$$

Feynman Rules – Weak

- In addition to g coupling factors for each vertex, from Chapter 1, real calcs also put other factors, some operators
 - from Lagrangian

Weak Force
Lagrangian term

$$(\bar{u}, \bar{c}, \bar{t}) \left[g_W \gamma^\mu (1 - \gamma^5) V \right] \begin{pmatrix} d \\ s \\ b \end{pmatrix} \rightarrow \begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix}$$

There is also a further weak “mixing” with ElectMag leading to writing g_w a diff way

Vector /Matrix (Dirac equation thing) operator
Also mixes parity:

- 1 : +parity,
- $-\gamma^5$ -parity!

For the weak force, this can be thought of additional coupling constant modifications

Neutrinos have the same thing

- “Nominal Couplings”

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}$$

- Actual couplings $\rightarrow \nu_1, \nu_2, \dots$ (ν_e, ν_μ , already the prime [ν'] states)
- P-NMS: Pontecorvo-Maki-Nakagawa-Sakata - neutrinos

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu1} & U_{\mu2} & U_{\mu3} \\ U_{\tau1} & U_{\tau2} & U_{\tau3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

$$u \rightarrow d + e^+ + \nu_e \quad \equiv \quad u \rightarrow d + e^+ + \nu_1 \quad + \quad u \rightarrow d + e^+ + \nu_2 \quad + \quad u \rightarrow d + e^+ + \nu_3$$

The coupling constants for the mass eigenstates are $U_{e1} \frac{g_W}{\sqrt{2}}$, $U_{e2} \frac{g_W}{\sqrt{2}}$, and $U_{e3} \frac{g_W}{\sqrt{2}}$.

- Leads of course to the oscillations:
- Time dependences determined by rest mass states ν_1, ν_2, ν_3
 - Usual QM phase oscillations for linear combinations:

$$|\nu_e\rangle = e^{-i\omega_1 t} |\nu_1\rangle + e^{-i\omega_2 t} |\nu_2\rangle + \dots$$

CKM Values → Wolfenstein

- CKM:
$$\begin{bmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{bmatrix} = \begin{bmatrix} 0.97370 & 0.2245 & 0.00382 \\ 0.221 & 0.987 & 0.0410 \\ 0.0080 & 0.0388 & 1.013 \end{bmatrix}.$$
- No underlying reason from Standard Model for CKM values—parameterizations are just empirical: (2022 vals)

- Two popular ways of parameterizing
- 1) “Wolfenstein → emphasizes diagonallness

$$\begin{pmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ \lambda & 1 - \frac{\lambda^2}{2} & A\lambda^3 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^3 & 1 \end{pmatrix} + O(\lambda^4)$$

complex → needed for CP violation

- Leptons/ ν 's : P-NMS: not as diagonal
 - Pontecorvo–Maki–Nakagawa–Sakata - neutrinos

$$|U| = \begin{bmatrix} |U_{e1}| & |U_{e2}| & |U_{e3}| \\ |U_{\mu 1}| & |U_{\mu 2}| & |U_{\mu 3}| \\ |U_{\tau 1}| & |U_{\tau 2}| & |U_{\tau 3}| \end{bmatrix} = \begin{bmatrix} 0.801 \dots 0.845 & 0.513 \dots 0.579 & 0.143 \dots 0.156 \\ 0.233 \dots 0.507 & 0.461 \dots 0.694 & 0.631 \dots 0.778 \\ 0.261 \dots 0.526 & 0.471 \dots 0.701 & 0.611 \dots 0.761 \end{bmatrix}$$

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OTHER FORM FOR
Mixing & CP violation:

$c_{12} \rightarrow \cos(\theta_{12})$
etc 3 angles and
a phase

$$\begin{bmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{13}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{13}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{13}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{13}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{13}} & c_{23}c_{13} \end{bmatrix}$$

Kaon Oscillations Ch 4

- Weak Interaction States $\neq$ Mass “Eigenstates”
 - (“What is a particle”?)
- Rather than oscillation (like ν 's) for quarks, always in bound states, but “oscillations” still seen e.g. in alternate decay modes

$$K_1 \rightarrow 2\pi, \quad K_2 \rightarrow 3\pi \quad \text{“Kshort” “Klong”} \quad (4.70)$$

Now, the 2π decay is much faster, because the energy released is greater. So if we start with a beam of K^0 's

$$|K^0\rangle = \left(\frac{1}{\sqrt{2}}\right) (|K_1\rangle + |K_2\rangle) \rightarrow \frac{1}{\sqrt{2}} [|\kappa_s\rangle + |\kappa_L\rangle] \quad (4.71)$$

$$|K_L^0\rangle = \frac{1}{\sqrt{1+|\varepsilon|^2}} (|K_2\rangle + \varepsilon |K_1\rangle)$$

very cool!

CKM Other Oscillations

- Weak Interaction States $\neq$ Mass “Eigenstates”
 - (“What is a particle”?)
- Many effects: E.g. any weak process that produces “one” weak flavor (e.g. “ ν_μ ”) produces mixed mass eigenstates ($\nu_\mu + \nu_e \dots$)
 - Neutrinos: “free particles” -- Time oscillation phase of “wave function components” go with mass eigenstates ... hence “neutrino oscillations”
 - For quarks, always in alternate decay modes

$$|K_L^0\rangle = \frac{1}{\sqrt{1+|\varepsilon|^2}} (|K_2\rangle + \varepsilon |K_1\rangle)$$

“oscillations” still seen e.g.

$$K_1 \rightarrow 2\pi, \quad K_2 \rightarrow 3\pi \quad (4.70)$$

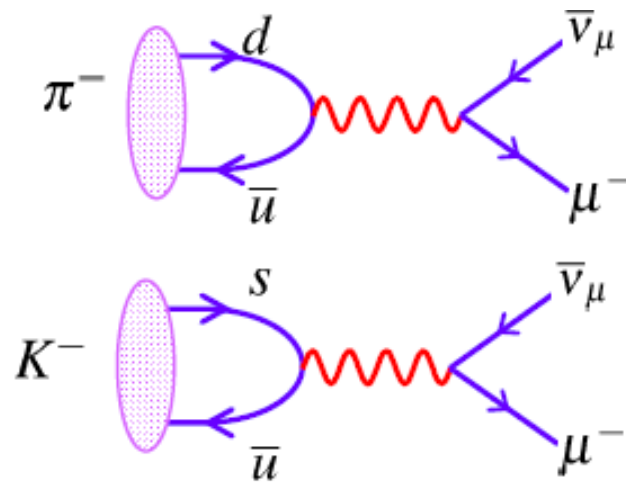
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$$|K^0\rangle = \left(\frac{1}{\sqrt{2}}\right) (|K_1\rangle + |K_2\rangle) \quad (4.71)$$

This section in Chapter 4: coolest in the book?

History: Cabbibo Angle

- E.g. to explain why π^- decay rate much $>$ than K^-
 - (OPPOSITE EXPECTATION FROM PHASE SPACE/ENERGY)



$$\cos \theta_c$$

$$\sin \theta_c$$

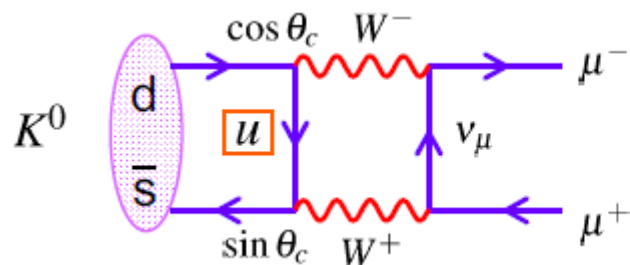
$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos \theta_c & \sin \theta_c \\ -\sin \theta_c & \cos \theta_c \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}$$

Mini CKM:

$$\theta_c = 13 \text{ degree} \\ (cC 0.97, sC 0.22)$$

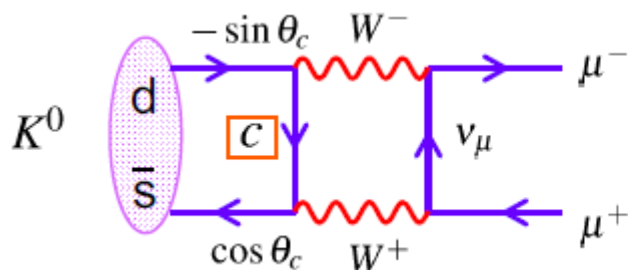
History: Cabbibo Angle

- Also explains $K_0 \rightarrow \mu\mu$ rate smaller than from leading diagram (historically this was first reason to introduce)
- Also $\rightarrow$ lead to first prediction of c quark existence !



$$M_1 \propto g_W^4 \cos \theta_c \sin \theta_c$$

- Historically, the observed branching was much smaller than predicted



$$M_2 \propto -g_W^4 \cos \theta_c \sin \theta_c$$

- Same final state so sum amplitudes

$$|M|^2 = |M_1 + M_2|^2 \approx 0$$

- Cancellation not exact because $m_u \neq m_c$

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos \theta_c & \sin \theta_c \\ -\sin \theta_c & \cos \theta_c \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}$$

“Rotation” CKM Unitary

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_2 & s_2 \\ 0 & -s_2 & c_2 \end{pmatrix} \begin{pmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\delta} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_3 & s_3 \\ 0 & -s_3 & c_3 \end{pmatrix}$$

Three rotations and a phase

Seminar Last Week

- I missed it $\rightarrow$ Super allowed?

Appendix I: Determination of the CKM Matrix

Non-examinable

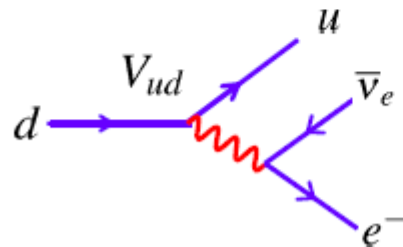
- The experimental determination of the **CKM matrix** elements comes mainly from measurements of leptonic decays (the leptonic part is well understood).
- It is easy to produce/observe meson decays, however theoretical uncertainties associated with the decays of bound states often limits the precision
- Contrast this with the measurements of the **PMNS matrix**, where there are few theoretical uncertainties and the experimental difficulties in dealing with neutrinos limits the precision.

①

$|V_{ud}|$

from nuclear beta decay

$$\begin{pmatrix} \times & \dots \\ \vdots & \vdots \\ \vdots & \vdots \end{pmatrix}$$



Super-allowed $0^+ \rightarrow 0^+$ beta decays are relatively free from theoretical uncertainties

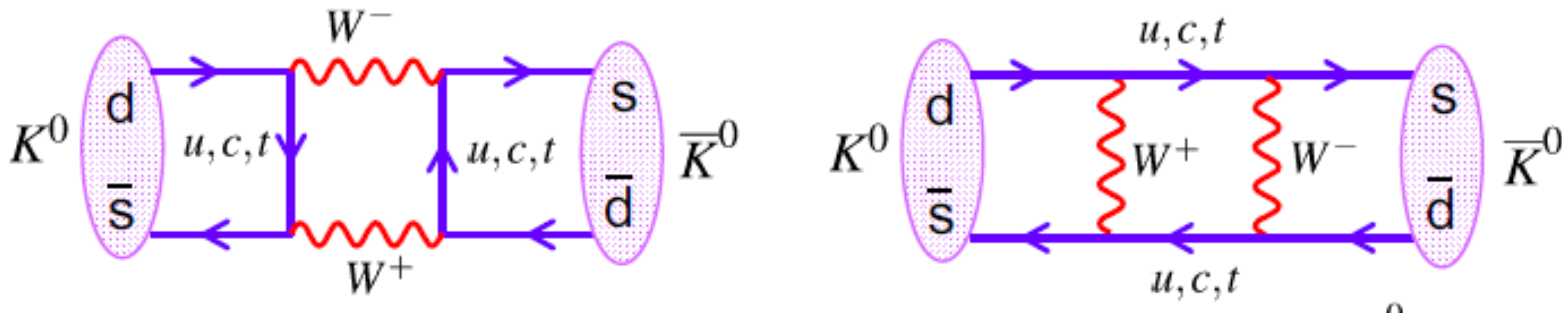
$$\Gamma \propto |V_{ud}|^2$$

$$|V_{ud}| = 0.97377 \pm 0.00027$$

$$(\approx \cos \theta_c)$$

CP Invariance $\rightarrow$ KO Oscillations

- KO – K0bar Oscillations possible
- (Notice $\rightarrow$ FLAVOR NON CONSERVATION)



- Kaons are pseudoscalars $P = -1^{(\ell+1)}$
- Charge Conjugation $\rightarrow$ K0 to K0bar

$$\begin{aligned} P|K^0\rangle &= -|K^0\rangle & P|\bar{K}^0\rangle &= -|\bar{K}^0\rangle & CP|K^0\rangle &= -|\bar{K}^0\rangle \\ C|K^0\rangle &= |\bar{K}^0\rangle & C|\bar{K}^0\rangle &= |K^0\rangle & CP|\bar{K}^0\rangle &= -|K^0\rangle \end{aligned}$$

Normalized eigenstates of CP are therefore:

$$\begin{aligned} |K_1\rangle &= \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle) & CP|K_1\rangle &= +|K_1\rangle \\ |K_2\rangle &= \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle) & CP|K_2\rangle &= -|K_2\rangle \end{aligned}$$

CP → Mixing to CP Violation

- $K_1(K_{\text{short}}) \rightarrow 2\pi$ $K_2(K_{\text{long}}) \rightarrow 3\pi$
- These states have approximately the same mass $m(K_S) \approx m(K_L) \approx 498 \text{ MeV}$
- But very different lifetimes: $\tau(K_S) = 0.9 \times 10^{-10} \text{ s}$ $\tau(K_L) = 0.5 \times 10^{-7} \text{ s}$
- Lifetime of $K_{\text{short}} \rightarrow$ similar to longer lived B mesons
 - (Nuclear lunch last week → use silicon/displaced vtx to ID in detectors)
 - (B^0 - B^0_{bar} also has mixing and is used in same context esp for CP violation)
 - DECAYS IN MIDDLE OF TYPICAL DETECTOR
- Lifetime of $K_L \rightarrow$ “stable” for like pion ($\tau \sim 10^{-8}$)
- Beam of K^0 becomes pure K_L within “a few cm” (Griff)
- Still find 1/1000 rate of $K^0 \rightarrow \pi^+\pi^- \rightarrow$ CP VIOLATION

$$|K_L^0\rangle = \frac{1}{\sqrt{1+|\epsilon|^2}} (|K_2\rangle + \epsilon |K_1\rangle)$$

(CP eigens)

$$K_L \rightarrow \pi^+ \pi^-$$

- actually two mechanisms – Indirect / Direct

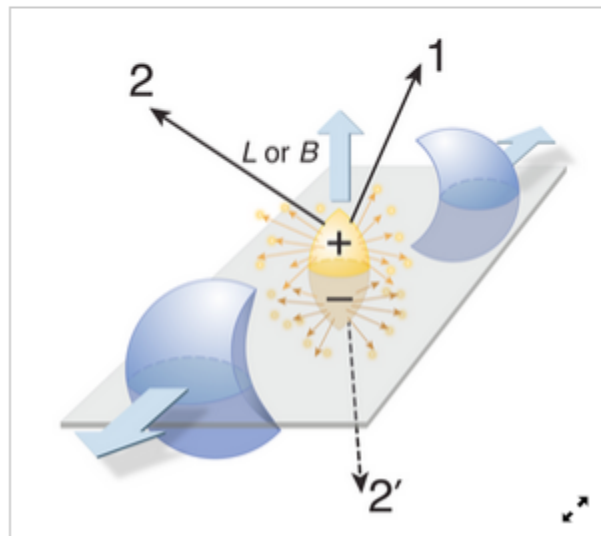
Looking for parity violation in heavy-ion collisions

Berndt Müller

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December 14, 2009 • *Physics 2*, 104

The STAR detector at RHIC has measured a signal that may indicate parity violation occurs in metastable regions of the superdense matter.



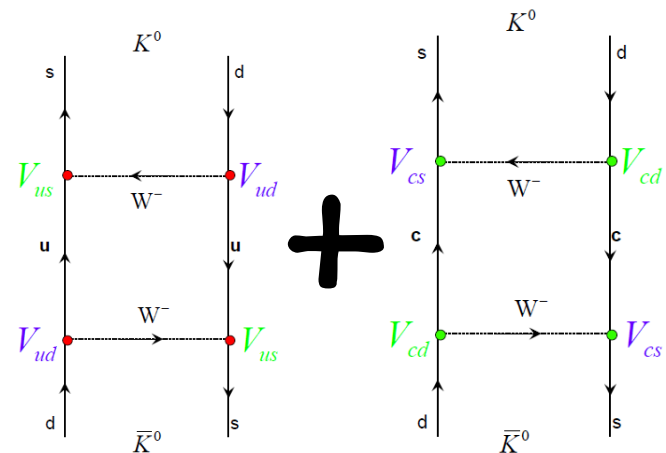
How CP Violation

- “Mixing” is needed but doesn’t cause CPVio a priori
- Need also complex phase $\rightarrow$ “from” the complex phase term of CKM and another process WITHOUT complex phase
- Due to Interference

$$\mathcal{M} = |\mathcal{M}_1| e^{i\theta_1} e^{i\phi_1} + |\mathcal{M}_2| e^{i\theta_2} e^{i\phi_2}$$

$$\bar{\mathcal{M}} = |\mathcal{M}_1| e^{i\theta_1} e^{-i\phi_1} + |\mathcal{M}_2| e^{i\theta_2} e^{-i\phi_2}$$

$$|\mathcal{M}|^2 - |\bar{\mathcal{M}}|^2 = -4 |\mathcal{M}_1| |\mathcal{M}_2| \sin(\theta_1 - \theta_2) \sin(\phi_1 - \phi_2).$$



Time dependence
 $\rightarrow$ oscillations $e^{i\omega t}$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_2 & s_2 \\ 0 & -s_2 & c_2 \end{pmatrix} \begin{pmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\delta} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_3 & s_3 \\ 0 & -s_3 & c_3 \end{pmatrix}$$

Three rotations and a phase

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{i\delta} \end{bmatrix} \begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-i\delta} \end{bmatrix} \begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{13}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{13}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{13}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{13}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{13}} & c_{23}c_{13} \end{bmatrix}.$$

The 2008 values for the standard parameters were:^[10]

$$\theta_{12} = 13.04^\circ \pm 0.05^\circ, \theta_{13} = 0.201^\circ \pm 0.011^\circ, \theta_{23} = 2.38^\circ \pm 0.06^\circ$$

and

$$\delta_{13} = 1.20 \pm 0.08 \text{ radians} = 68.8^\circ \pm 4.5^\circ.$$

from Wikipedia CKM

Strong CP Problem/ Symm violations in Strong

CP seems conserved in Strong Interaction
-but this may be a problem for some
→ why should it be, if electroweak is not?
STRONG CP Problem → “fine tuning” ?!?

Parity/CP violation searched for in strong interactions

IN HEAVY ION COLLISIONS

“Local” CP vio in Strong

Low E Nonperturb QCD
Complicated

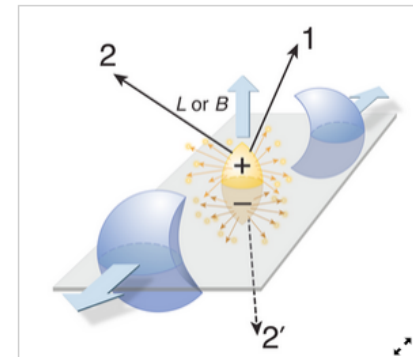
Looking for parity violation in heavy-ion collisions

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The STAR detector at RHIC has measured a signal that may indicate parity violation occurs in metastable regions of the superdense matter.



Total Conservation Table

	Q	L	B	J	P	G	C	CP	T	CPT
Strong	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Electromagnetic	✓	✓	✓	✓	✓	X	✓	✓	✓	✓
Weak	✓	✓	✓	✓	X	X	X	X	X	✓

Q = electric charge, L = lepton number, B = baryon number, J = spin,
P = parity, G = G-parity, C = charge conjugation, T = time reversal

CP, CPT are products

Not included....quark flavour...which is conserved by the strong and electromagnetic interaction and by the neutral weak interaction but not by the charged weak interaction. Also, could add isospin I and third component of isospin I_3 (also, did not include cons. of energy etc).

CPT MUST BE CONSERVED (why? locality and Lorentz inv of QFT) $\rightarrow$ T vio $\leftrightarrow$ CP vio*

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Weak	✓	✓	✓	✓	X	X	X	X	X	✓

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CPT MUST BE CONSERVED (why? locality (i.e. anti-nonlocal) and Lorentz inv of QFT)*

T vio $\leftrightarrow$ CP vio

Baryon /Lepton Number

- Look at Wikipedia: Mathm. Formulation of Standard Model

Summary of Conservation Laws

	Q	L	B	J	P	G	C	CP	T	CPT
Strong	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Electromagnetic	✓	✓	✓	✓	✓	X	✓	✓	✓	✓
Weak	✓	✓	✓	✓	X	X	X	X	X	✓

ry
ations

Perturbative QFT and the interaction picture
Lagrangian formalism
Detailed information
Physics beyond the Standard Model
See also
References and external links

Bibcode:2012CRPhy..13..566M. doi:10.1016/j.crhy.2012.04.008. S2CID 119272967.

12. The baryon number in SM is only conserved at the classical level. There are non-perturbative effects that do not conserve baryon number: [Baryon Number Violation, report prepared for the Community Planning Study – Snowmass 2013](#)

13. The lepton number in SM is only conserved at the classical level. There are non-perturbative effects that do not conserve lepton number: see Fuentes-Martín, J.; Portolés, J.; Ruiz-Femenía, P. (January 2015). "Instanton-mediated baryon number violation in non-universal gauge extended models". *Journal of High Energy Physics*. **2015** (1): 134. arXiv:1411.2471. Bibcode:2015JHEP...01..134F. doi:10.1007/JHEP01(2015)134. ISSN 1029-8479. or [Baryon and lepton numbers in particle physics beyond the standard model](#)

14. The violation of lepton number and baryon number cancel each other out and in effect $B - L$ is an exact symmetry of the Standard Model. Extension of the Standard Model with massive Majorana neutrinos breaks B-L symmetry, but extension with massive Dirac neutrinos does not: see Ma, Ernest; Srivastava, Rahul (2015-08-30). "Dirac or inverse seesaw neutrino masses from gauged B-L symmetry". *Modern Physics Letters A*. **30** (26): 1530020. arXiv:1504.00111. Bibcode:2015MPLA...3030020M. doi:10.1142/S0217732315300207. ISSN 0217-7323. S2CID 11911538. Heeck, Julian (December

called Accidental Symm of SM
full lagrangian
Lepton number (L) same
Neutrino oscillations explicity
break L number consv.
Therefore B is expected to be
correspondingly non-consv b.c.
SM Lagrange has even strong
L-B number conservation
constraint

Baryon Number implemented as:

$$B = \frac{1}{3}(n - \bar{n})$$

ie this U(1) current again:

Of course, there's the ordering ambiguity here, but we'll press on and insert the mode expansion to yield

$$\hat{Q}_N = \frac{1}{2} \int d^3p \left(-\hat{a}_p^\dagger \hat{a}_p + \hat{b}_p \hat{b}_p^\dagger - \hat{a}_p \hat{a}_p^\dagger + \hat{b}_p^\dagger \hat{b}_p \right). \quad (12.13)$$

Finally, we normal order to give a conserved charge of

$$N[\hat{Q}_N] = \int d^3p \left(\hat{b}_p^\dagger \hat{b}_p - \hat{a}_p^\dagger \hat{a}_p \right). \quad (12.14)$$